

## Details:

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## Title

Smart Autonomous Vacuum Cleaning Agent For Hospitals

## Aim

To design and implement a Smart Autonomous Vacuum Cleaning Agent using Python that can detect dirty locations, move between different locations, clean them, monitor its battery level, automatically move to a charging station when the battery is low, and calculate its performance measure.

## Experiment Objectives

To design functionality for a vacuum agent that:

1. Detects clean and dirty locations in an environment.
2. Automatically cleans dirty locations.
3. Moves from one location to another as required.
4. Maintains a performance measure based on its actions.
5. Monitors the battery level of the vacuum cleaner.
6. Moves to a predefined charging station when the battery becomes low.
7. Recharges the battery and continues the cleaning process.
8. Demonstrates the working of a Simple Reflex Agent in a real-world environment such as a hospital.

## Theory

An AI agent is a system that perceives its environment through sensors and performs actions using actuators to achieve a particular goal.

A Simple Reflex Agent selects an action based on the current state of the environment using predefined condition-action rules.

For the proposed vacuum cleaning agent:

- 0 represents a Clean location.
- 1 represents a Dirty location.

The environment consists of multiple locations such as:

- Reception
- ICU
- Ward
- Corridor
- Pharmacy
- Waiting Area

The agent randomly starts at one of these locations.

The agent follows rules such as:

Rule 1: If the current location is dirty, clean it.

Rule 2: If another location is dirty, move to that location.

Rule 3: If the battery level is less than or equal to 30%, move to the charging station.

Rule 4: After charging, resume the cleaning operation.

The performance measure is calculated using rewards and penalties.

Action -> Performance

Cleaning a dirty location -> +10

Moving to another location -> -1

Moving to charging station -> -1

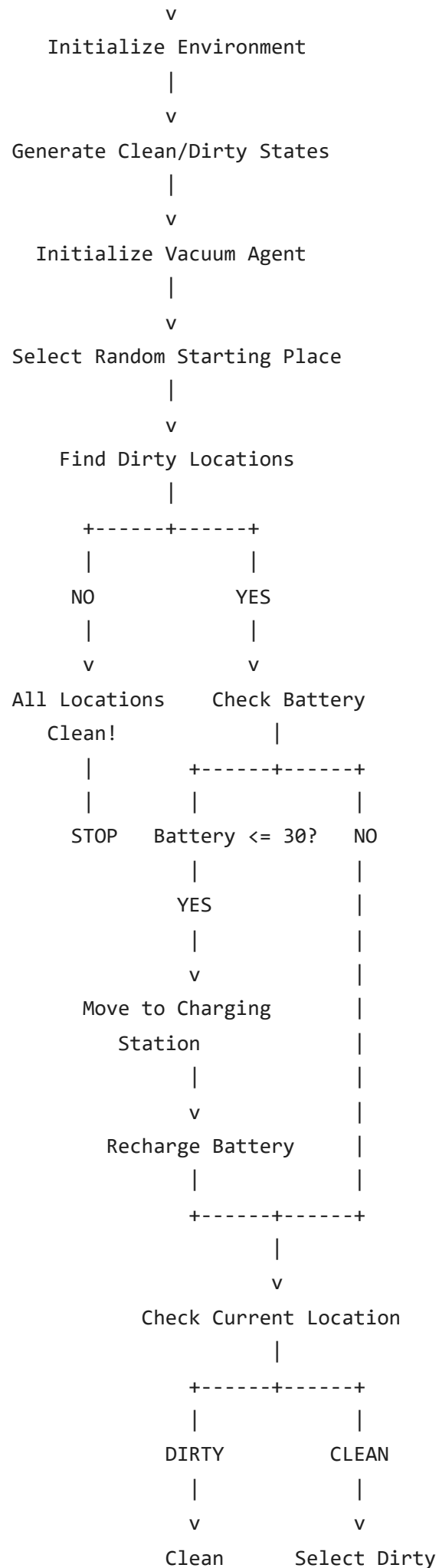
Successful charging -> +2

Thus, the agent attempts to clean all locations while minimizing unnecessary movement and managing its battery.

## Flowchart Details

START

|



```

|           Location
|           |
|           v
|           Move
|           |
+-----+-----+
|
|           v
|         Clean
|           |
|           v
|       Update Score
|           |
|           v
|   Find Dirty Locations
|           |
|           v
|       REPEAT

```

## Code

```

import random

class Env:
    def __init__(self):
        self.locationConditions={
            "Reception": random.randint(0, 1),
            "ICU": random.randint(0, 1),
            "Ward": random.randint(0, 1),
            "Corridor": random.randint(0, 1),
            "Pharmacy": random.randint(0, 1),
            "Waiting Area": random.randint(0,1)
        }

class vaccumCleaner:
    def __init__(self,env):
        self.env=env
        self.currLocation=random.choice(list(self.env.locationConditions.keys()))

        self.battery=100
        self.score=0
        self.movements=0
        self.chargingCount=0
        print("Initial conditions: ",env.locationConditions)
        print("Vaccum Cleaner starts at :",self.currLocation)
        print("Battery: ",self.battery,"%")
        self.run_cleaner()

```

```
def find_dirty_locations(self):
    dirtyLoc=[]
    for location,condition in self.env.locationConditions.items():
        if(condition==1):
            dirtyLoc.append(location)
    return dirtyLoc

def move(self,location):
    self.score-=1
    self.battery-=5
    self.movements+=1
    print("\nMoving from ",self.currLocation," to ",location)
    self.currLocation=location
    print("Battery: ",self.battery,"%")

def clean(self):
    self.score+=10
    self.battery-=20
    print("\nCleaning ",self.currLocation,...)
    self.env.locationConditions[self.currLocation]=0
    print("Cleaning completed")
    print("Battery: ",self.battery,"%")

def charge(self):
    print("\nBattery is low!")
    print("Moving to charging station")
    self.score-=1
    self.battery-=5
    self.movements+=1

    print("Battery charged successfully!")
    self.score+=2
    self.battery=100
    self.chargingCount+=1
    print("Battery: ",self.battery,"%")

def final_disp(self):
    print("\n\nFinal Results:")
    print("Environment: ",self.env.locationConditions)
    print("Current position: ",self.currLocation)
    print("Battery: ",self.battery)
    print("Performance measure: ",self.score)
    print("Number of movements: ",self.movements)
    print("Number of charging operations: ",self.chargingCount)

def run_cleaner(self):
    if(self.env.locationConditions[self.currLocation]==1):
        self.clean()
    dirtyLocations=self.find_dirty_locations()
    if(len(dirtyLocations)==0):
        print("\nAll locations are clean!")
    else:
        for loc in dirtyLocations:
            if(self.battery<=30):
                self.charge()
```

```
        if(self.currLocation==loc):
            continue
        else:
            self.move(loc)
            self.clean()
    self.final_disp()
```

```
Testenv=Env()
```

```
TestVaccumCleaner=vaccumCleaner(Testenv)
```

```
Initial conditions: {'Reception': 1, 'ICU': 1, 'Ward': 0, 'Corridor': 1, 'Pharmacy': 0}
Vaccum Cleaner starts at : ICU
```

```
Battery: 100 %
```

```
Cleaning ICU ...
```

```
Cleaning completed
```

```
Battery: 80 %
```

```
Moving from ICU to Reception
```

```
Battery: 75 %
```

```
Cleaning Reception ...
```

```
Cleaning completed
```

```
Battery: 55 %
```

```
Moving from Reception to Corridor
```

```
Battery: 50 %
```

```
Cleaning Corridor ...
```

```
Cleaning completed
```

```
Battery: 30 %
```

```
Battery is low!
```

```
Moving to charging station
```

```
Battery charged successfully!
```

```
Battery: 100 %
```

```
Moving from Corridor to Pharmacy
```

```
Battery: 95 %
```

```
Cleaning Pharmacy ...
```

```
Cleaning completed
```

```
Battery: 75 %
```

```
Final Results:
```

```
Environment: {'Reception': 0, 'ICU': 0, 'Ward': 0, 'Corridor': 0, 'Pharmacy': 1}
```

```
Current position: Pharmacy
```

```
Battery: 75
```

```
Performance measure: 38
```

```
Number of movements: 4
```

```
Number of charging operations: 1
```

# Experiment Outcome

The experiment demonstrates the implementation of a Simple Reflex Vacuum Agent using Python.

The agent is able to:

- Detect dirty locations.
- Clean dirty locations.
- Move between locations.
- Monitor its battery.
- Charge when the battery is low.
- Maintain a performance score.
- Display the final cleaning status and performance statistics.

## Result Analysis

The Smart Autonomous Vacuum Cleaning Agent was successfully implemented.

The agent randomly generated the cleanliness conditions of different hospital locations and selected a random starting location. It successfully identified dirty locations and cleaned them.

The agent also maintained a performance score. Cleaning a location increased the score, while movement and charging introduced a small penalty. The battery level was continuously monitored, and when it reached the specified threshold, the agent performed the charging operation before continuing its cleaning task.

## Conclusion

The Smart Autonomous Vacuum Cleaning Agent was successfully designed and implemented using Python. The agent demonstrates the concept of a Simple Reflex Agent by taking actions based on the current condition of its environment.

The proposed system extends the traditional two-location vacuum problem by introducing multiple locations, autonomous movement, battery monitoring, charging functionality, and performance measurement. The same concept can be further extended to real-world applications such as hospital cleaning, college cleaning, office cleaning, and parking-area cleaning.

The experiment provides a practical understanding of how AI agents perceive an environment, make decisions based on predefined rules, perform actions, and

evaluate their performance.

## Github link:

[click](#)

## Colab Link

[click](#)